THESIS RESEARCH SYSTEM

Compile once. Extract indefinitely.

TWIX 2 learns a reusable structured-data extractor from a small sample of a template-based PDF family, then ships ordinary Python code for every later page.

ID_05 CASE STUDY

95.5%

HELD-OUT ACCURACY

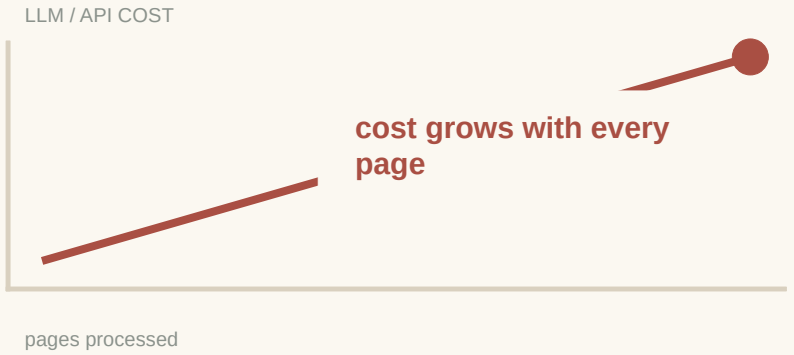
15 test pages
56 matched records
224 physical blocks
0 runtime model calls

1 September 2026

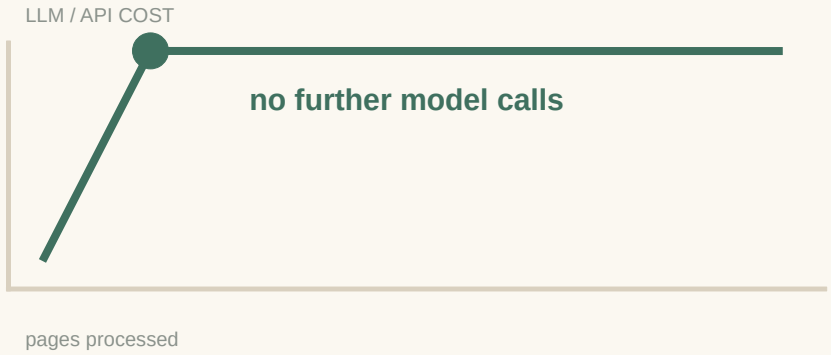
The problem is not one page - it is the repeated model call

A frontier model can parse a page well. The operational cost still grows every time another page is sent.

CONVENTIONAL MODEL-PER-PAGE



TWIX 2 COMPILE-ONCE



The hypothesis: structure turns a broad reasoning task into smaller code-generation tasks that cheaper - and potentially local - models can solve.

02 / END-TO-END PIPELINE

Models compile the extractor; ordinary Python runs it

Learning happens once; extraction becomes a repeatable, model-free runtime.

ONE-TIME COMPILATION

1 **Detect + split**
Find content start; create train and test OCR.

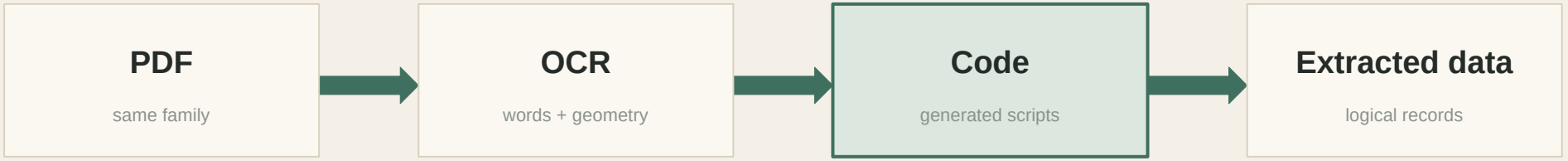
2 **Infer template**
Define node ids, fields and metadata patterns.

3 **Write code**
Generate separator and one extractor per node.

4 **Train gate**
Evaluate all train pages; repair if needed.

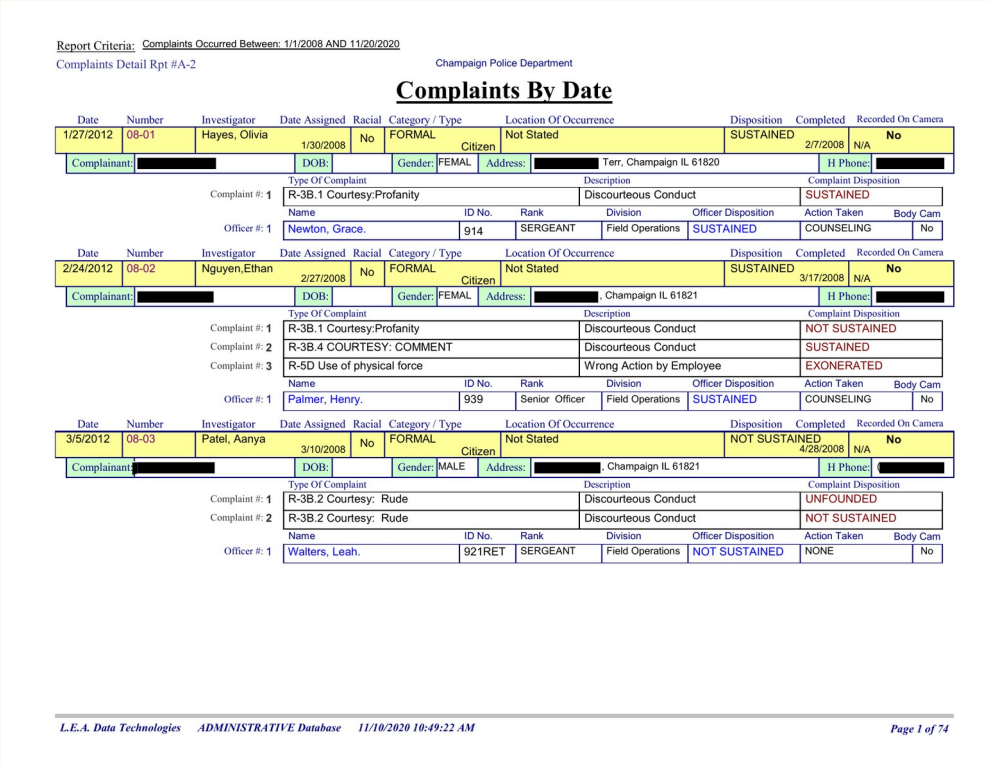
5 **Package**
Copy model-free code into pipeline/.

REPEATED RUNTIME



A 20-page sample contains a stable four-block record pattern

The report repeats complaint summaries, complainant details, complaint rows and officer rows.



Report Criteria: Complaints Occurred Between: 1/1/2008 AND 11/20/2020

Complaints Detail Rpt #A-2

Champaign Police Department

Complaints By Date

Date	Number	Investigator	Date Assigned	Racial	Category / Type	Location Of Occurrence	Disposition	Completed	Recorded On Camera
1/27/2012	08-01	Hayes, Olivia	1/30/2008	No	FORMAL Citizen	Not Stated	SUSTAINED	2/7/2008	N/A No

Complainant: [redacted]	DOB:	Gender: FEMAL	Address: [redacted] Terr, Champaign IL 61820	H Phone: [redacted]

	Type Of Complaint	Description	Complaint Disposition
Complaint #: 1	R-3B.1 Courtesy:Profanity	Discourteous Conduct	SUSTAINED

	Name	ID No.	Rank	Division	Officer Disposition	Action Taken	Body Cam
Officer #: 1	Newton, Grace.	914	SERGEANT	Field Operations	SUSTAINED	COUNSELING	No

Date	Number	Investigator	Date Assigned	Racial	Category / Type	Location Of Occurrence	Disposition	Completed	Recorded On Camera
2/24/2012	08-02	Nguyen,Ethan	2/27/2008	No	FORMAL Citizen	Not Stated	SUSTAINED	3/17/2008	N/A No

Complainant: [redacted]	DOB:	Gender: FEMAL	Address: [redacted], Champaign IL 61821	H Phone: [redacted]

	Type Of Complaint	Description	Complaint Disposition
Complaint #: 1	R-3B.1 Courtesy:Profanity	Discourteous Conduct	NOT SUSTAINED
Complaint #: 2	R-3B.4 COURTESY: COMMENT	Discourteous Conduct	SUSTAINED
Complaint #: 3	R-5D Use of physical force	Wrong Action by Employee	EXONERATED

	Name	ID No.	Rank	Division	Officer Disposition	Action Taken	Body Cam
Officer #: 1	Palmer, Henry.	939	Senior Officer	Field Operations	SUSTAINED	COUNSELING	No

Date	Number	Investigator	Date Assigned	Racial	Category / Type	Location Of Occurrence	Disposition	Completed	Recorded On Camera
3/5/2012	08-03	Patel, Aanya	3/10/2008	No	FORMAL Citizen	Not Stated	NOT SUSTAINED	4/28/2008	N/A No

Complainant: [redacted]	DOB:	Gender: MALE	Address: [redacted], Champaign IL 61821	H Phone: [redacted]

	Type Of Complaint	Description	Complaint Disposition
Complaint #: 1	R-3B.2 Courtesy: Rude	Discourteous Conduct	UNFOUNDED
Complaint #: 2	R-3B.2 Courtesy: Rude	Discourteous Conduct	NOT SUSTAINED

	Name	ID No.	Rank	Division	Officer Disposition	Action Taken	Body Cam
Officer #: 1	Walters, Leah.	921RET	SERGEANT	Field Operations	NOT SUSTAINED	NONE	No

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20

SOURCE PAGES

5 train + 15 test

7,270

OCR WORDS

global ids + geometry

4

TEMPLATE NODES

3 tables + 1 KV

56

TEST RECORDS

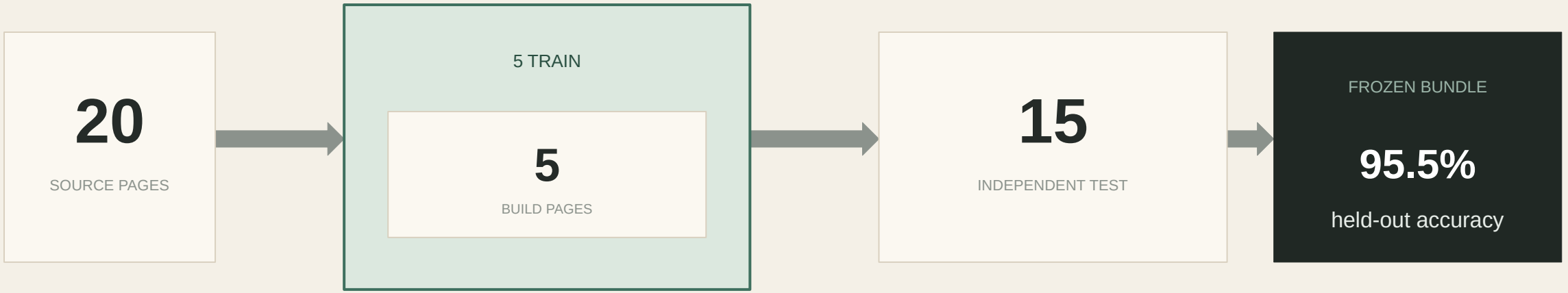
after deterministic grouping

Source redactions are preserved in every visual.

04 / INDEPENDENT EVALUATION SPLIT

Five pages compile the extractor; fifteen pages test it

This run uses all five training pages for code generation, then evaluates the frozen bundle on fifteen independent pages.



Result

The bundle is generated once from five training pages, then evaluated without changing its code on fifteen independent pages.

Four node ids define every lookup, block and extractor

Field names describe semantics; bid describes visual position; metadata regexes distinguish repeated page furniture from record content.

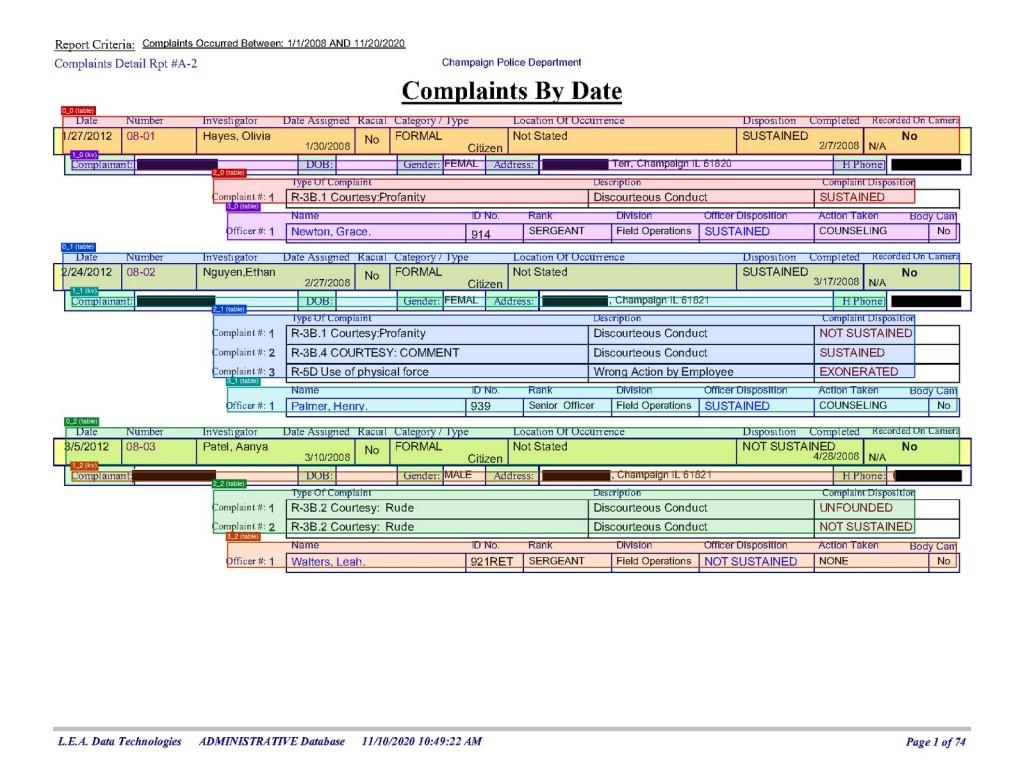

```
{
  "node_id": 2,
  "type": "table",
  "fields": [
    "Type Of Complaint",
    "Description",
    "Complaint Disposition"
  ],
  "bid": [2],
  "complexity": {
    "same_row_structure": true,
    "nested": false
  }
}
```


NODE	TYPE	SIZE	ROLE
0	**table**	10 fields	Complaint summary
1	**key-value**	5 fields	Complainant details
2	**table**	3 fields	Complaint rows
3	**table**	7 fields	Officer rows

9 metadata regexes: 7 document-level + 2 record-level

Generated geometry code finds page-local blocks

On page 1, twelve visual regions map to four node types; multi-page tables remain separate continuation blocks.



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IDENTITY CONTRACT

block_id = <node_id>_<instance>

0_1 summary
1_1 complainant
2_1 complaints
3_1 officers

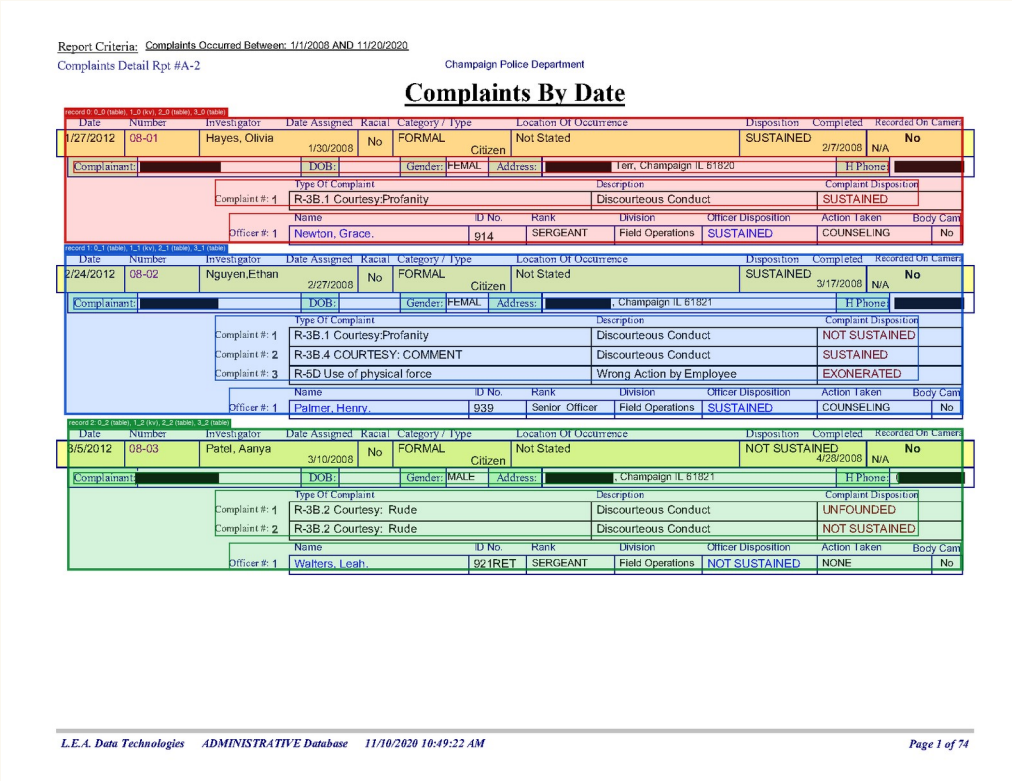
uid = physical join key

Page-local only

A block emitted under page P may contain only page-P words.

Blocks 0 → 1 → 2 → 3 form one complaint record

Deterministic grouping preserves every physical block while producing a record-level JSON structure.



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ONE RECORD

- **0_0** summary
- **1_0** KV details
- **2_0** complaint rows
- **3_0** officer rows

56 / 56

matched test records

Routing reserves the stronger model for difficult nodes

Nodes 0 and 1 use Qwen3.8 Flash; the harder table extractors use GLM-5.3 Flash.

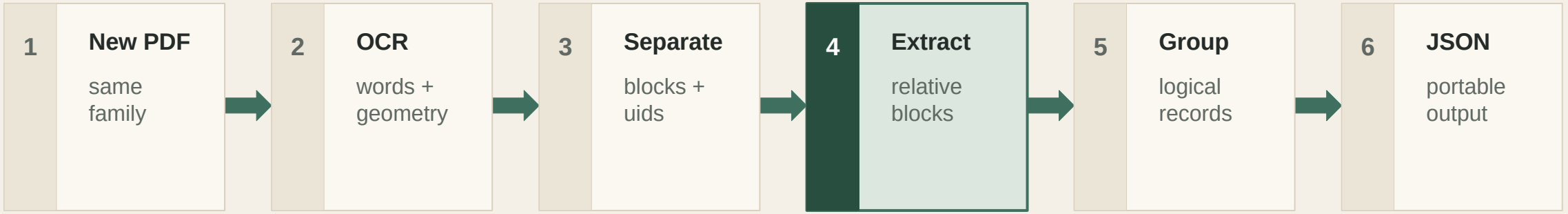
NODE 0	TABLE	198 lines	Qwen3.8 Flash
NODE 1	KEY-VALUE	139 lines	Qwen3.8 Flash
NODE 2	TABLE	224 lines	GLM-5.3 Flash
NODE 3	TABLE	190 lines	GLM-5.3 Flash

```
# extraction_scripts/<node_id>.py
def extract(block: dict) -> dict:
    ...


# runtime dispatch
node_id = block["block_id"].rsplit("_", 1)[0]
result =
extractor[node_id].extract(relative_block)
result["page"] = page
result["uid"] = uid
```

The deployed path is a six-step local program

LangGraph, OpenCode and model providers disappear from the production extraction path.



NO LANGGRAPH **NO OPENCODE** **NO MODEL CALLS**

The marginal LLM/API cost of extraction is zero after compilation; local compute remains.

Physical blocks stay visible inside each logical record

Grouping preserves page, uid and block identity while making the output usable downstream.


```
{
  "record_id": "0",
  "blocks": [
    {"block_id": "0_0", "type": "table", "page": 1},
    {"block_id": "1_0", "type": "kv",    "page": 1},
    {"block_id": "2_0", "type": "table", "page": 1},
    {"block_id": "3_0", "type": "table", "page": 1}
  ]
}
```


RECORD 0 / SELECTED VALUES

DATE	NUMBER
7/19/2012	**08-14**
INVESTIGATOR	DISPOSITION
Kim, Joon	**UNFOUNDED**
GENDER	OFFICER
MALE	**Carter, Felix.**
COMPLAINT	ACTION
R-3B.3 Courtesy: Conduct	**NONE**

Held-out extraction reaches 95.5% accuracy

Across 15 independent pages, every record and node type is found; remaining errors are localized value-boundary and column-assignment issues.

95.1%

TRAIN ACCURACY

5 pages / 19 records

95.5%

TEST ACCURACY

15 pages / 56 records

1.000

RECORD MATCH F1

perfect precision + recall

1.000

NODE MATCH F1

typed precision + recall



The residual gap is narrow and actionable

The scorer reports field-level reasons that can be handed directly to the optional repair loop.

TYPICAL MISMATCH

Gold	**FORMAL**
Candidate	**FORMAL Citizen**
Reason	A neighboring column value crosses the Category / Type boundary.

WHAT IS NOT FAILING

- No missing gold records
- No extra candidate records
- No missing or extra nodes
- No node type mismatches

Every phase leaves a file that can be inspected or deployed

The run directory is simultaneously a provenance record, a visual QA workspace and a portable extractor.

UI/runs/run-20260901-184959-d47392/

├── ocr_extraction_full.csv	full-document OCR
├── template.json	schema + metadata contract
├── metadata.json	models, configuration, scores
├── artifacts/	block + record overlays
├── agent_space/	train/test PDFs and OCR
└── **pipeline/**	deployable bundle + outputs

pipeline/

pipeline.py

block_separation.py

extraction_scripts/

record_grouping.py

requirements.txt

template.json

grouped.json

pipeline_health.json

eval.json + eval.txt

One compilation cost is amortized over every later page

The architectural claim is about marginal model cost, not about eliminating local compute.

AFTER COMPILATION

$0 **marginal LLM/API cost**

- No model context window grows with production pages.
- The same code handles each same-family page.

ID_05 COMPILATION SCALE

3.40M **input + output tokens**

One-time code-generation workload

This expense is paid once for the document family, then amortized across every future extraction.

Local OCR, CPU, storage and hosting still consume resources.

Use models to write the extractor - not every extraction

id_05 preserves record structure and approaches frontier-level content quality on an independent split.

ID_05 RESULT

5	15	56	95.5%
training pages	test pages	matched records	overall accuracy

0 runtime model calls

A reusable program replaces a repeated model call.

BOUNDARIES

- Family-specific, not a generic PDF extractor.
- Layout drift and OCR errors may require recompilation.
- Ground truth is only as reliable as its reference process.
- Zero model cost still requires local compute.